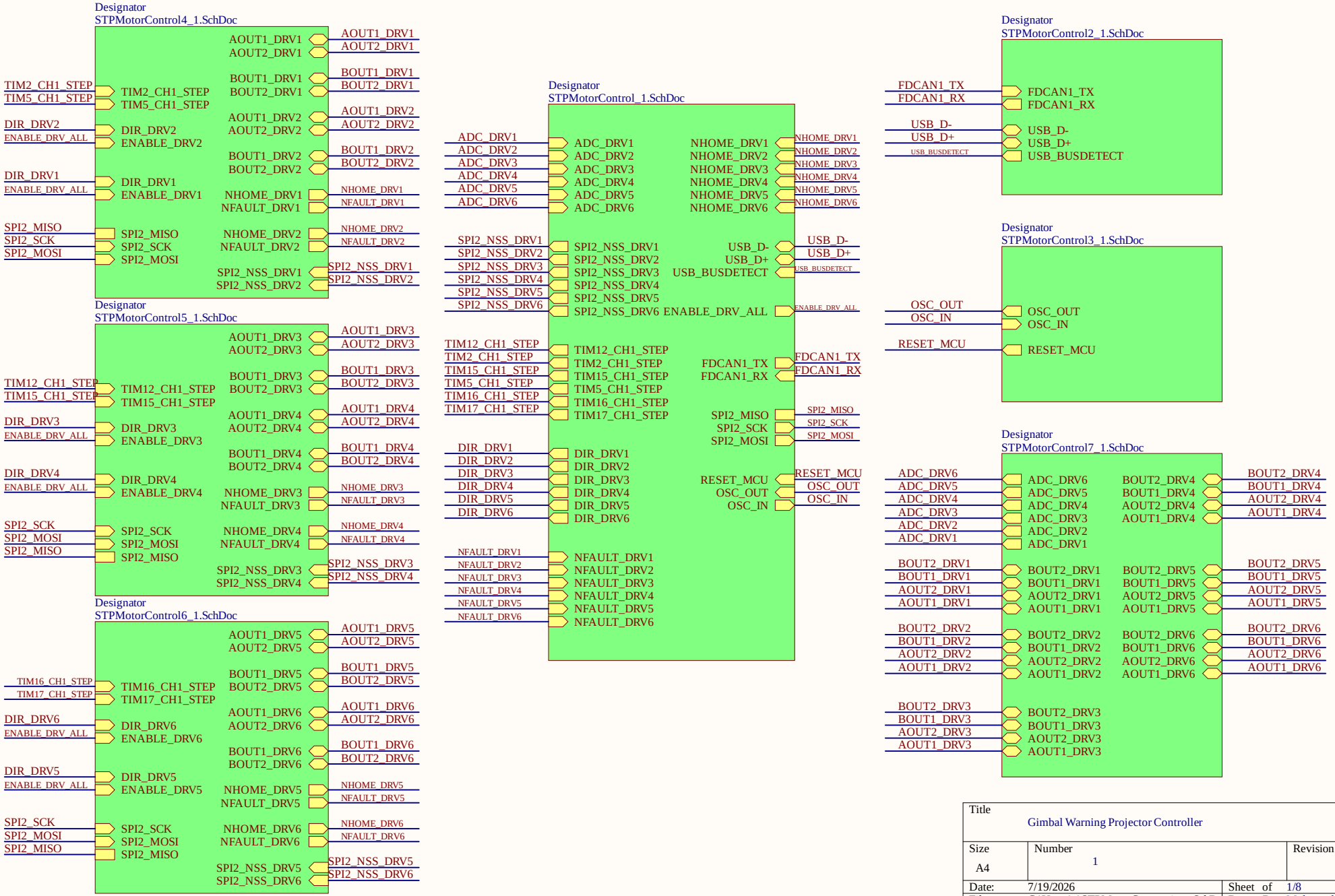
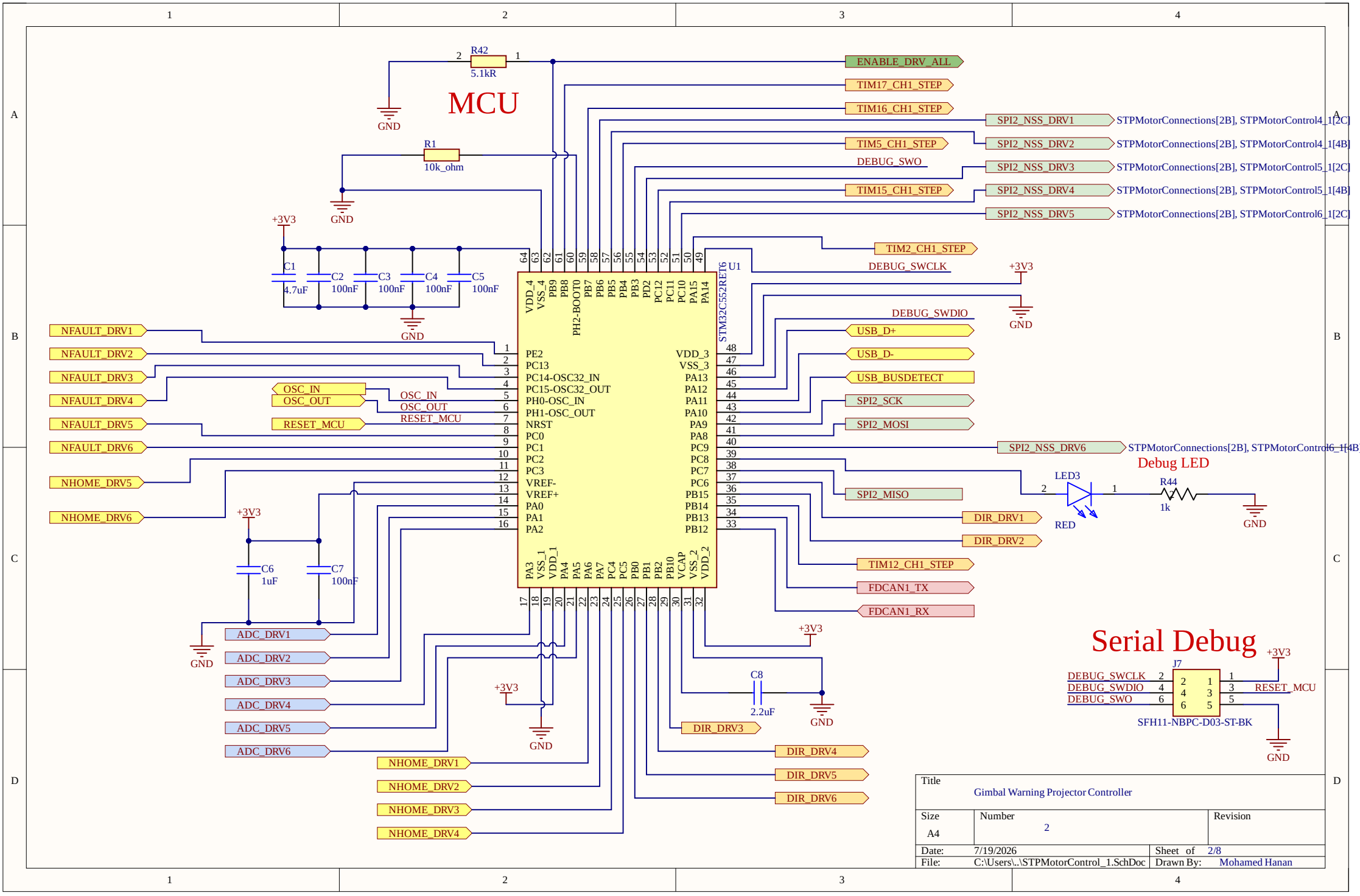
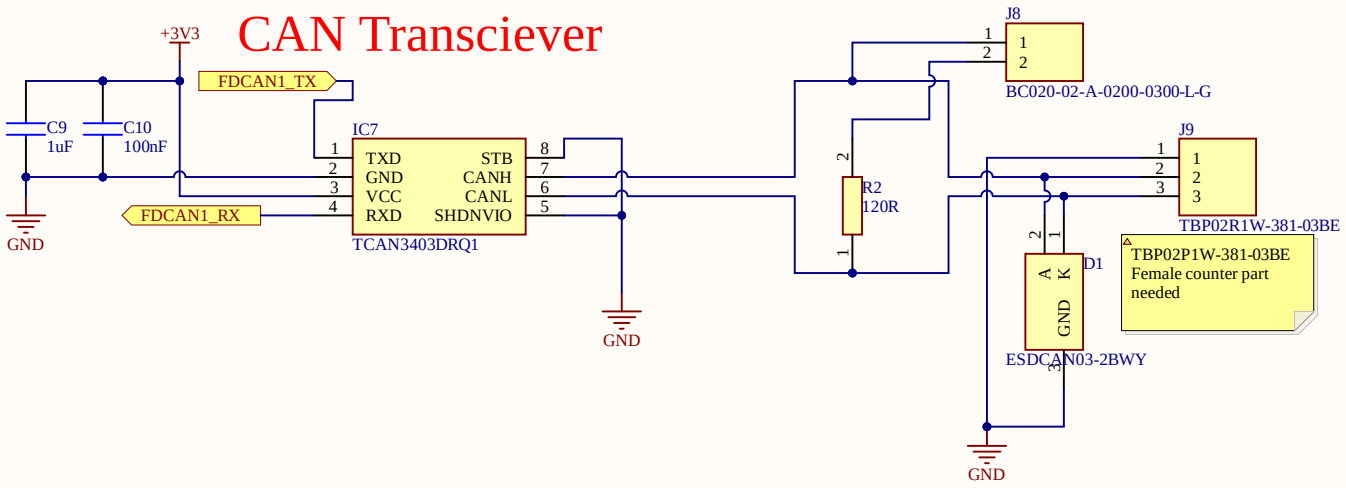
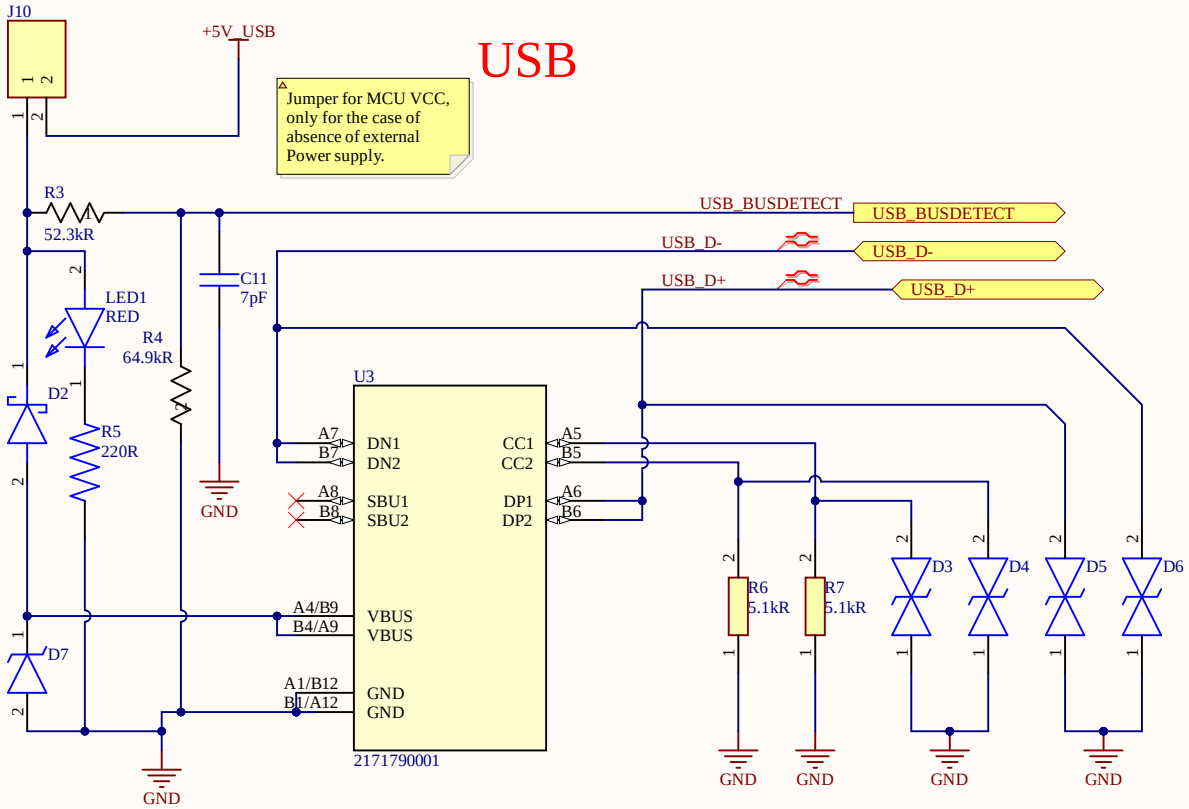


Title	Gimbal Warning Projector Controller	
Size: A4	Number: 1	Revision:
Date: 7/19/2026		Sheet of 1/8
File: C:\Users\..\STPMotorConnections.SchDoc		Drawn By: Mohamed Hanan

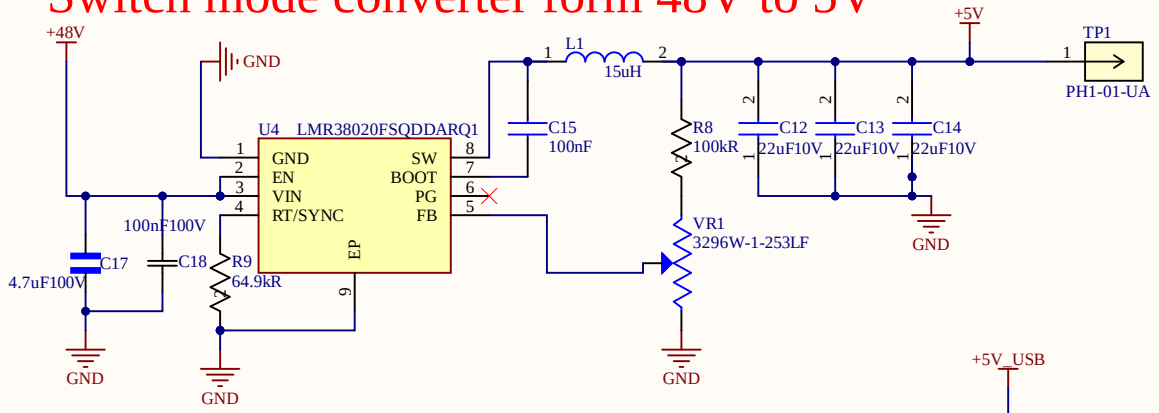

MCU
R42
5.1kR
R1
10k_ohm
GND
+3V3
C1
4.7uF
C2
100nF
C3
100nF
C4
100nF
C5
100nF
U1
STM32C552RET6
ENABLE_DRV_ALL
TIM17_CH1_STEP
TIM16_CH1_STEP
SPI2_NSS_DRV1
STPMotorConnections[2B], STPMotorControl4_1[2C]
TIM5_CH1_STEP
SPI2_NSS_DRV2
STPMotorConnections[2B], STPMotorControl4_1[4B]
DEBUG_SWO
SPI2_NSS_DRV3
STPMotorConnections[2B], STPMotorControl5_1[2C]
TIM15_CH1_STEP
SPI2_NSS_DRV4
STPMotorConnections[2B], STPMotorControl5_1[4B]
SPI2_NSS_DRV5
STPMotorConnections[2B], STPMotorControl6_1[2C]
TIM2_CH1_STEP
DEBUG_SWCLK
DEBUG_SWDIO
USB_D+
USB_D-
USB_BUSDETECT
SPI2_SCK
SPI2_MOSI
SPI2_NSS_DRV6
STPMotorConnections[2B], STPMotorControl6_1[4B]
SPI2_MISO
DIR_DRV1
DIR_DRV2
TIM12_CH1_STEP
FDCAN1_TX
FDCAN1_RX
NFAULT_DRV1
NFAULT_DRV2
NFAULT_DRV3
NFAULT_DRV4
NFAULT_DRV5
NFAULT_DRV6
NHOME_DRV5
NHOME_DRV6
OSC_IN
OSC_OUT
RESET_MCU
C6
1uF
C7
100nF
C8
2.2uF
ADC_DRV1
ADC_DRV2
ADC_DRV3
ADC_DRV4
ADC_DRV5
ADC_DRV6
NHOME_DRV1
NHOME_DRV2
NHOME_DRV3
NHOME_DRV4
DIR_DRV3
DIR_DRV4
DIR_DRV5
DIR_DRV6
Debug LED
LED3
RED
R44
1k
Serial Debug
J7
SFH11-NBPC-D03-ST-BK
Title
Gimbal Warning Projector Controller
Size
A4
Number
2
Revision
Date: 7/19/2026
Sheet of 2/8
File: C:\Users\..\STPMotorControl_1.SchDoc
Drawn By: Mohamed Hanan



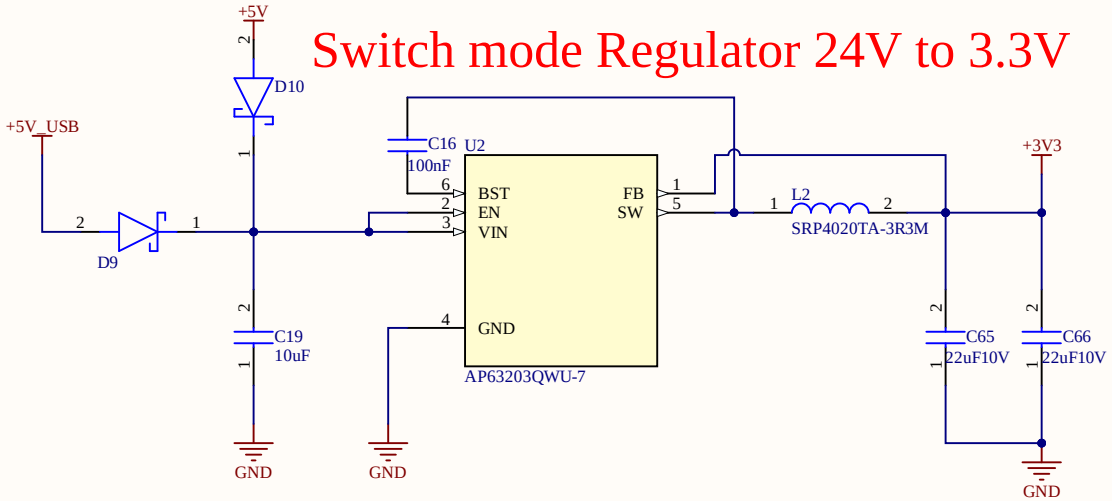


Title	Gimbal Warning Projector Controller	
Size: A4	Number: 3	Revision:
Date: 7/19/2026	Sheet of 3/8	
File: C:\Users\..\STPMotorControl2_1.SchDoc	Drawn By: Mohamed Hanan	

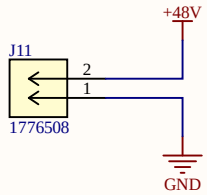
Switch mode converter form 48V to 5V



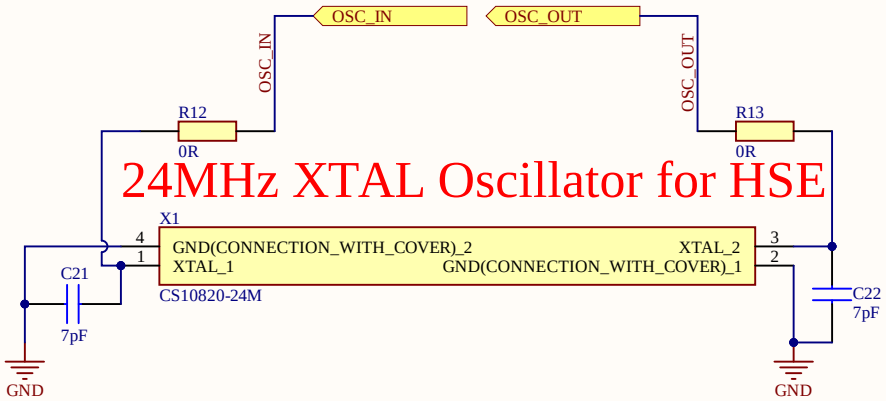
Switch mode Regulator 24V to 3.3V



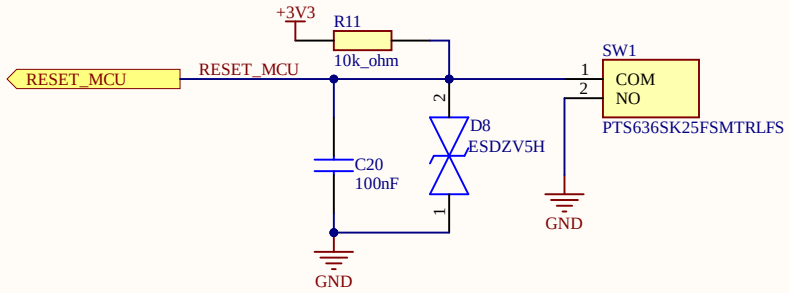
48V input from main source



24MHz XTAL Oscillator for HSE

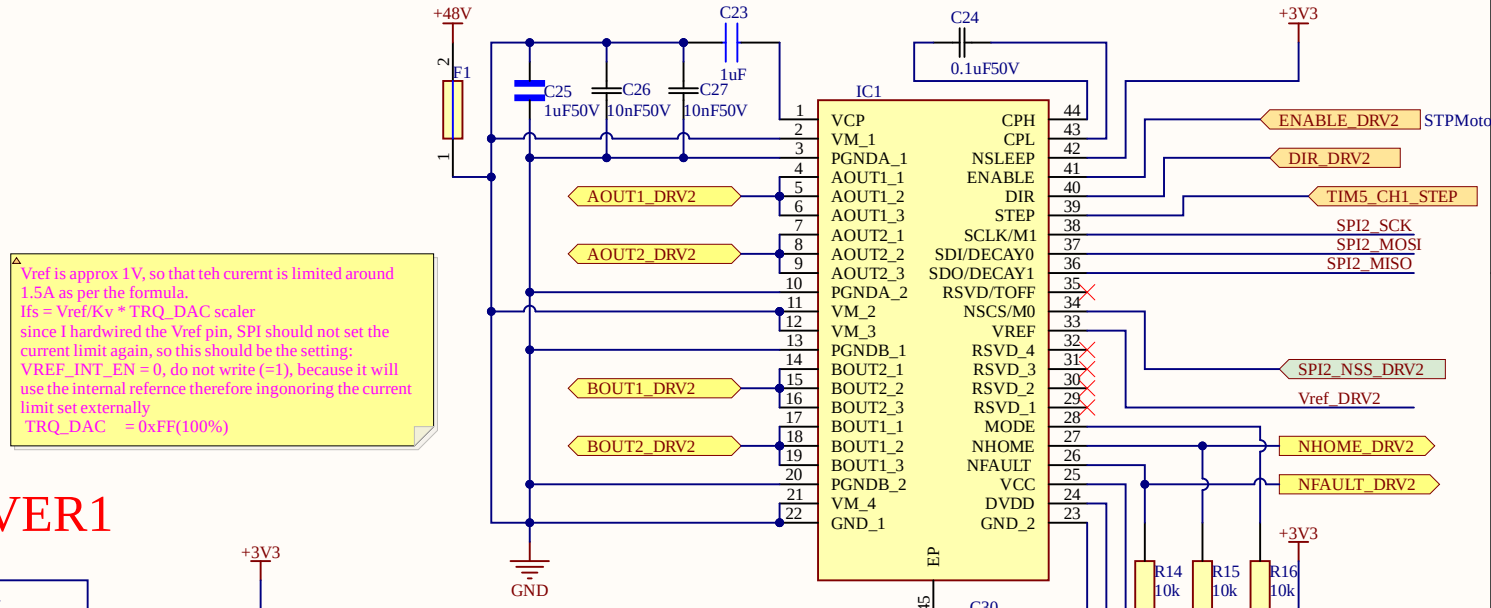


Switch 4 MCU RESET PIN



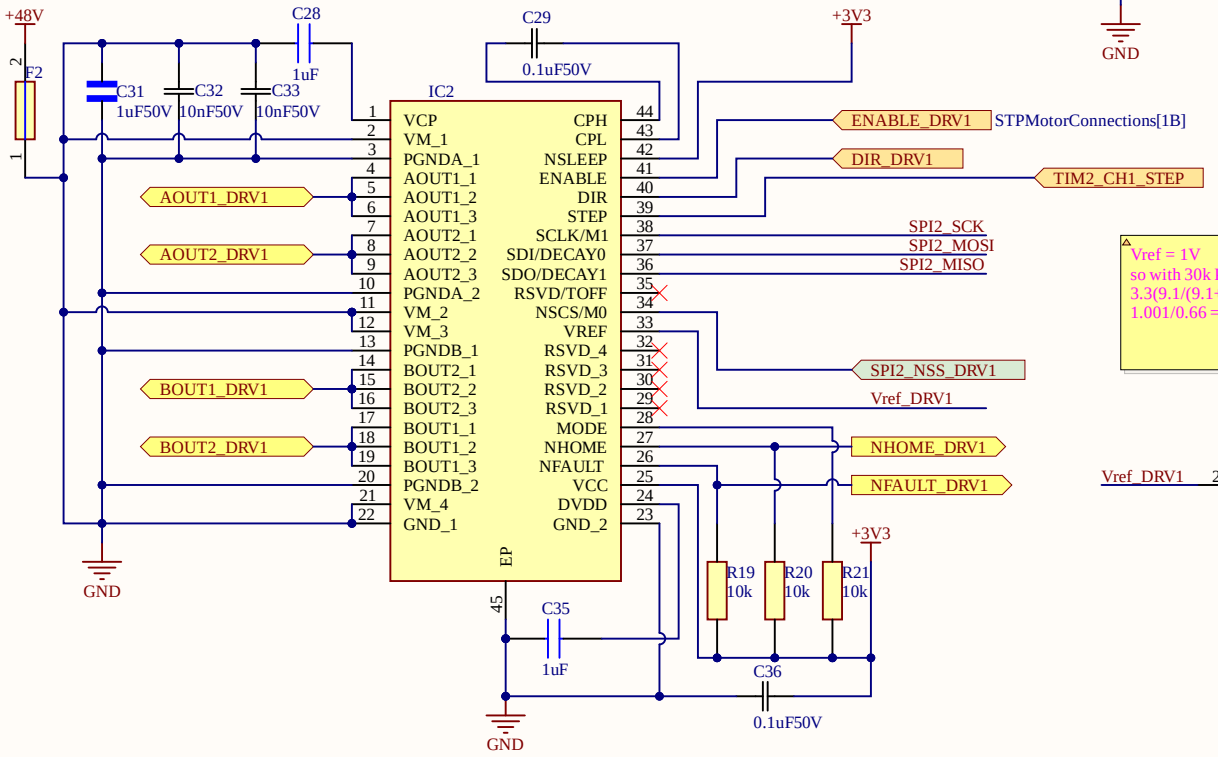
Title: Gimbal Warning Projector Controller		
Size: A4	Number: 4	Revision:
Date: 7/19/2026	Sheet of: 4/8	
File: C:\Users\..\STPMotorControl3_1.SchDoc	Drawn By: Mohamed Hanan	

STEPPER DRIVER2

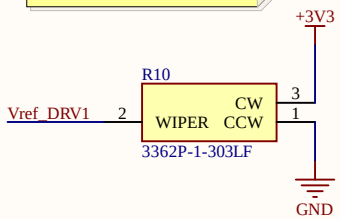


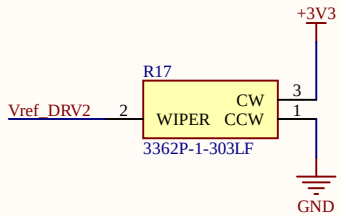
Vref is approx 1V, so that teh curernt is limited around 1.5A as per the formula.
Ifs = Vref/Kv * TRQ_DAC scaler
since I hardwired the Vref pin, SPI should not set the current limit again, so this should be the setting:
VREF_INT_EN = 0, do not write (=1), because it will use the internal refernce therefore ingonoring the current limit set externally
TRQ_DAC = 0xFF(100%)

STEPPER DRIVER1



Vref = 1V
so with 30k POT
3.3(9.1/(9.1+20.9)) = 1.001
1.001/0.66 = 1.51A

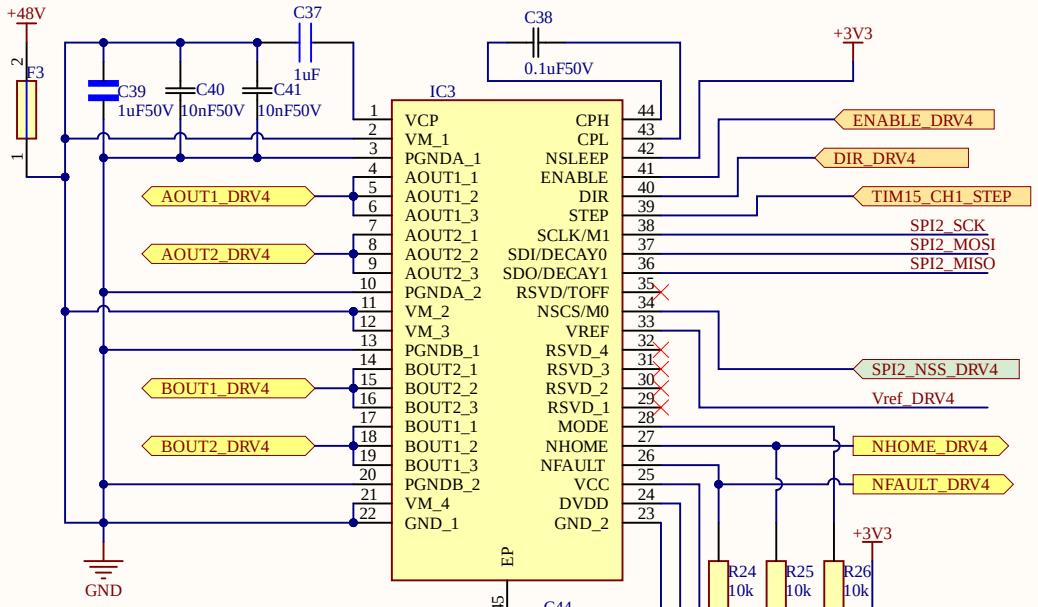




SPI2_SCK
SPI2_MOSI
SPI2_MISO

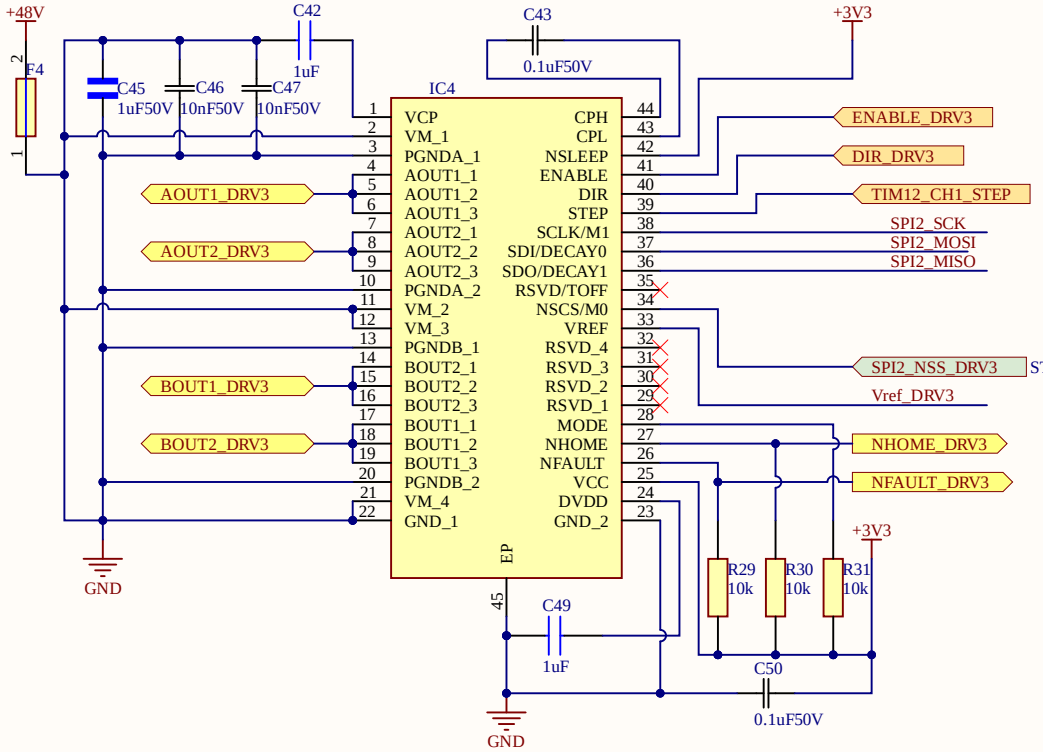
Title	Gimbal Warning Projector Controller		
Size: A4	Number: 5		Revision:
Date: 7/19/2026		Sheet of 5/8	
File: C:\Users\..\STPMotorControl4_1.SchDoc		Drawn By: Mohamed Hanan	

STEPPER DRIVER4

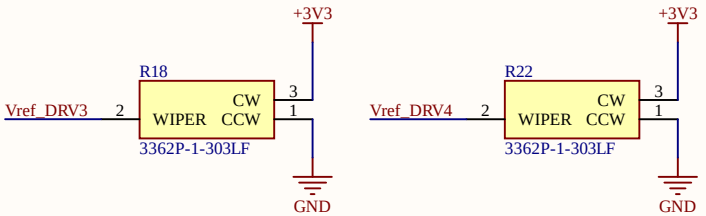


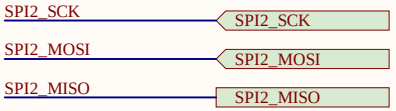
Vref is approx 1V, so that teh curernt is limited around 1.5A as per the formula.
Ifs = Vref/Kv * TRQ_DAC scaler
since I hardwired the Vref pin, SPI should not set the current limit again, so this should be the setting:
VREF_INT_EN = 0, do not write (=1), because it will use the internal refernce therefore ingonoring the current limit set externally
TRQ_DAC = 0xFF(100%)

STEPPER DRIVER3



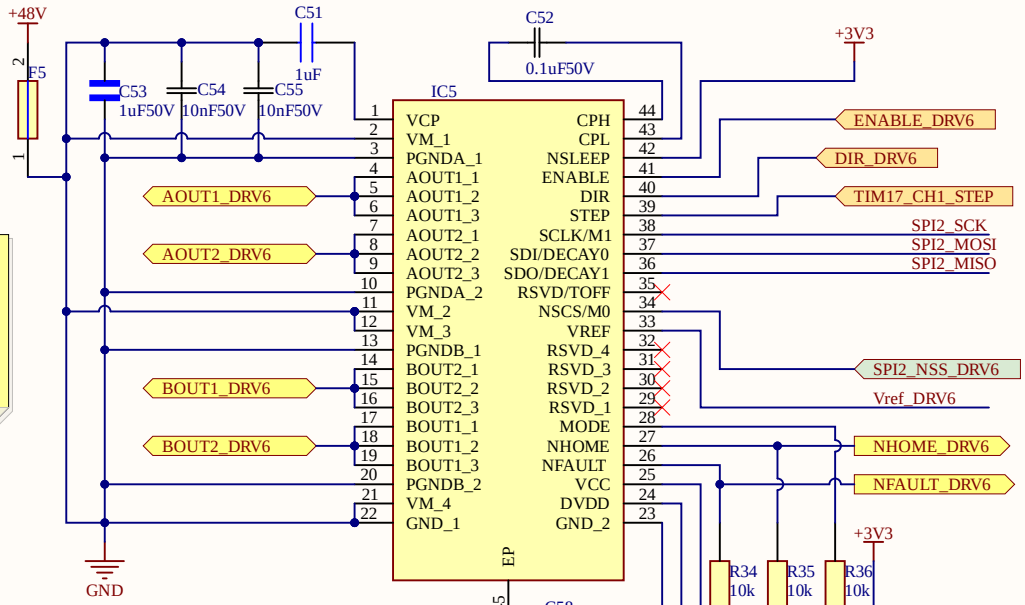
Vref = 1V
so with 30k POT
3.3(9.1/(9.1+20.9)) = 1.001
1.001/0.66 = 1.51A





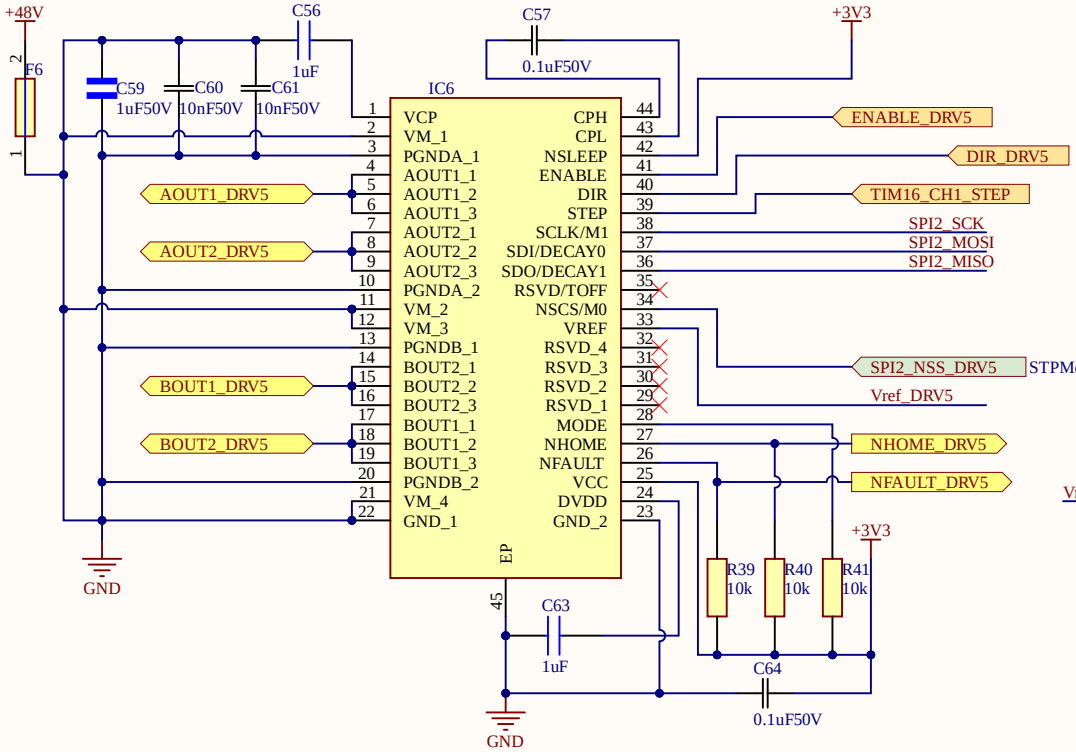
Title: Gimbal Warning Projector Controller		
Size: A4	Number: 6	Revision:
Date: 7/19/2026	Sheet of: 6/8	
File: C:\Users\..\STPMotorControl5_1.SchDoc	Drawn By: Mohamed Hanan	

STEPPER DRIVER6

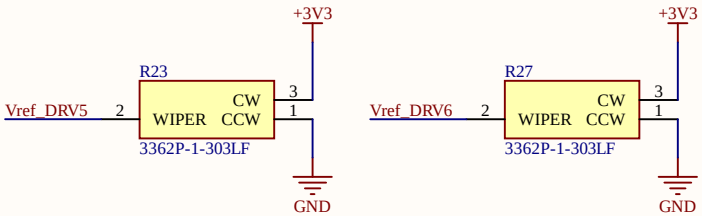


Vref is approx 1V, so that teh curernt is limited around 1.5A as per the formula.
Ifs = Vref/Kv * TRQ_DAC scaler
since I hardwired the Vref pin, SPI should not set the current limit again, so this should be the setting:
VREF_INT_EN = 0, do not write (=1), because it will use the internal refernce therefore ingonoring the current limit set externally
TRQ_DAC = 0xFF(100%)

STEPPER DRIVER5

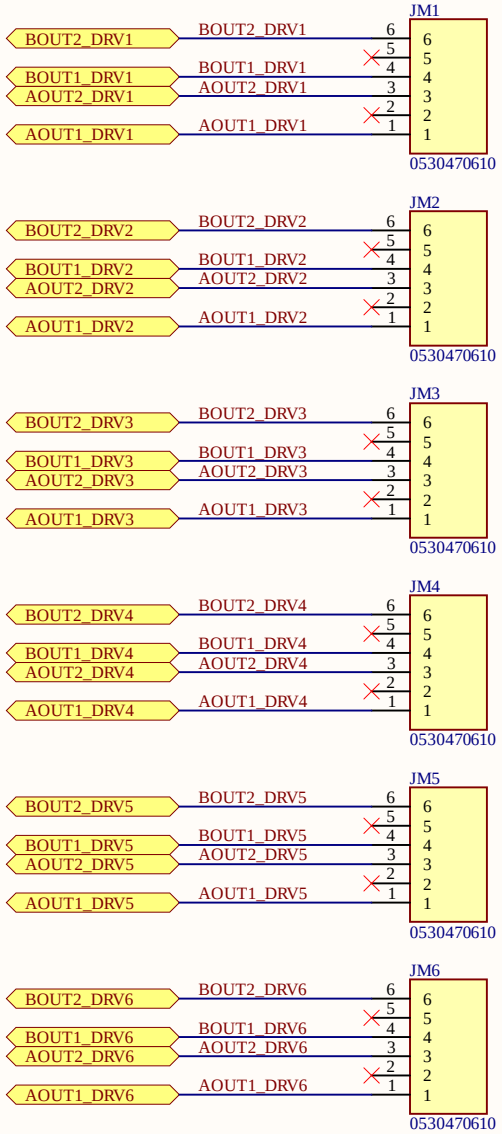


Vref = 1V
so with 30k POT
3.3(9.1/(9.1+20.9)) = 1.001
1.001/0.66 = 1.51A

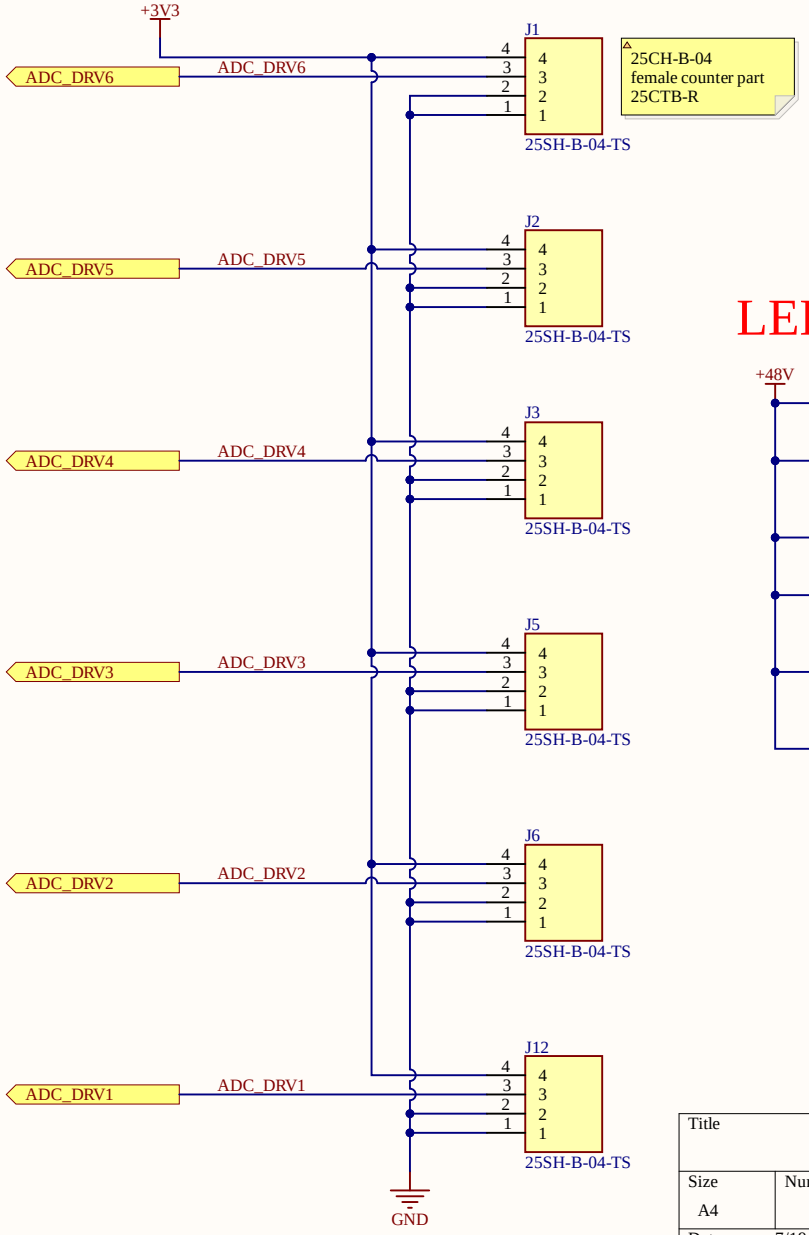


Title: Gimbal Warning Projector Controller		
Size: A4	Number: 7	Revision:
Date: 7/19/2026	Sheet of: 7/8	
File: C:\Users\..\STPMotorControl6_1.SchDoc	Drawn By: Mohamed Hanan	

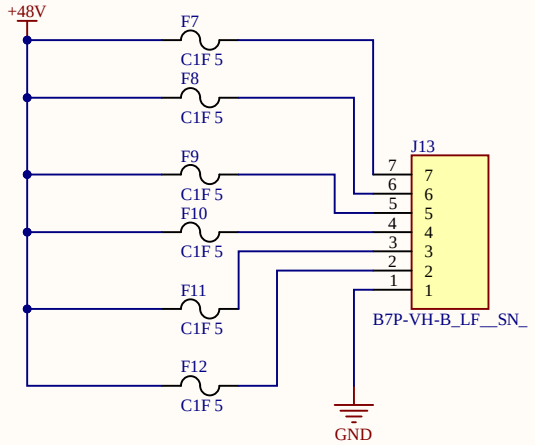
STEPPER MOTOR CONNECTOR



MAGNETIC ENCODER CONNECTOR



LED modules Connector



Title	Gimbal Warning Projector Controller		
Size A4	Number 8		Revision
Date: 7/19/2026		Sheet of	8/8
File: C:\Users\..\STPMotorControl7_1.SchDoc		Drawn By:	Mohamed Hanan

Comment	Description	Designator	Footprint	LibRef	Quantity
4.7uF	Multilayer Ceramic Capacitors MLCC - SMD/SMT	C1	CAPC1608X90N	CL10A475KP8NNNC	1
100nF	Samsung Electro-Mechanics CL05B104KO5NNNC 100nF Multilayer Ceramic Capacitor MLCC 16Vdc +/-10% Tolerance SMD	C2, C3, C4, C5, C7, C10, C15, C16, C20	CAPC1005X55N	CL05B104KO5NNNC	9
1uF	Samsung Electro-Mechanics 0402 (1005M) 1uF Multilayer Ceramic Capacitor MLCC 16Vdc +/-10% SMD CL05A105KO5NNNC	C6, C9, C23, C28, C30, C35, C37, C42, C44, C49, C51, C56, C58, C63	CAPC1005X55N	CL05A105KO5NNNC	14
2.2uF	Samsung Electro-Mechanics 0603 CL 2.2uF Ceramic Multilayer Capacitor, 10 V, +85C, X5R Dielectric, +/-10% SMD	C8	CAPC1608X90N	CL10A225KP8NNNC	1
7pF	CAPCER7PF50V C0G/NP0 0603	C11, C21, C22	CAPC1608X90N	CL10C070CB8NNNC	3
22uF10V	Capacitor	C12, C13, C14, C55, C66	GRM18x	GRM188R61A226ME15D	5
4.7uF100V	SMD capacitor X7S(EIA) with capacitance 4.7uF Tol 20%. Rated voltage 100Vdc 125C	C17	CAPC3216X190N	GRM31CC72A475ME11L	1
100nF100V	Description	C18	CAP_0603_VAL	0603B104K101CT	1
10uF	Capacitor	C19	CAPC3216X180N	CL31A106MAHNNNE	1
0.1uF50V	Description	C24, C29, C34, C36, C38, C43, C46, C50, C52, C57, C62, C64	CAP_GCM1555_0402_MLR	GCM155R71H104KE02J	12
1uF50V	SMD capacitor X5R(EIA) with capacitance 1.0uF Tol 20%. Rated voltage 50Vdc 85C	C25, C31, C39, C45, C53, C59	CAPC1005X60N	GRM155R61H105ME05D	6
10nF50V	SMD Comm X7R, Ceramic, 0.01 uF, 10%, 50 VDC, X7R, SMD, MLCC, Temperature Stable, Class II, 0603	C26, C27, C32, C33, C40, C41, C46, C47, C54, C55, C60, C61	CAPC17595_87N_KEM	C0603C103K5RACTU	12
ESDCAN03-2BWY	TVS Diode (Uni-directional)	D1	SOT65P210X110-3N	ESDCAN03-2BWY	1
VS-30BQ015-M3_9AT	Schottky Diode	D2, D9, D10	DKM795592/62N	VS-30BQ015-M3_9AT	3
ESD2V5H-1BLI2	Ultra low clamping single line bidirectional ESD protection	D3, D4, D5, D6	ESD2V5H1BLI2	ESD2V5H-1BLI2	4
ESDA25P35-1U1M	TVS Diode (Uni-directional)	D7	ESDA25P351U1M	ESDA25P35-1U1M	1
ESD2V5H	Ultra low clamping single line bidirectional ESD protection	D8	ESD2V5H1BLI2	ESD2V5H-1BLI2	1
0685T1000-01	Fuse	F1, F2, F3, F4, F5, F6	FUSC3216X83N	0685T1000-01	6
C1F5	Description	F7, F8, F9, F10, F11, F12	C1_BEL	C1F5	6
DRV8462DDWR	Integrated Circuit	IC1, IC2, IC3, IC4, IC5, IC6	SOP64P810X120-45N	DRV8462DDWR	6
TCAN3403DRQ1	Integrated Circuit	IC7	SOIC127P600X175-8N - duplicate	TCAN3403DRQ1	1
2SSH-B-04-TS	Connector	J1, J2, J3, J5, J6, J12	2SSH-B04TS	2SSH-B-04-TS	6
SFH11-NBPC-D03-ST-BK	Connector	J7	SFH11NBPCD03STBK	SFH11-NBPC-D03-ST-BK	1
BCD20-02-A-0200-0300-L-G	Connector	J8, J10	BCD2002A020003001LG	BCD20-02-A-0200-0300-L-G	2
TBP02R1W-381-03BE	Connector	J9	RHDRRA3MB0P0XB11_1X3_2182X820X760H	TBP02R1W-381-03BE	1
1776508	PCB headers, nominal current: 12 A, rated voltage (III/2): 320 V, number of positions: 2, pitch: 5.08 mm, color: green, contact surface: Tin, mounting: Wave soldering, Gold-plated contact surface / color: Green RAL 6021	J11	CONN_1776508	1776508	1
B7P-VH-B_LF__SN_	Connector	J13	B7PVHBLFSN	B7P-VH-B_LF__SN_	1
0530470610	Connector	JM1, JM2, JM3, JM4, JM5, JM6	530470610	0530470610	6
15uH	Inductor	L1	1217AS-H100M-P3	1217AS-H-150M-P3	1
SRR0207A-3R9M	Inductor	L2	SRR0207A3R9M	SRR0207A-3R9M	1
RED	LED	LED1, LED3	LEDC1608X55N	LTST-C191KRKT	2
10k_ohm	YAGEO (FH/COMF) RC0603 F, 10k OHM, 0.1W	R1, R11	RESC1608X55N	RC0603FR0710KL	2
120R	Resistor	R2	RCS1206_E3	RCS120612ORFKEA	1
52.3kR	SMD RES52.3K-1/16W-1%-0402	R3	RES_RC0402_YAG	RC0402FR-0752K3L	1
64.9kR	Description	R4, R9	RC0603N_YAG	RC0603FR-0764K9L	2
220R	Fixed Resistor CRCQ 0603 220RF5%	R5	RESC1608X55N	CRCCQ0603JE20R	1
5.1kR	Resistor	R6, R7, R42	RESC1608X55N	RC0603FR-075K1L	3
100kR	0.1/CRCW0603 100 100K1%ET1 e3	R8	RES_CRCW_0603_VIS	CRCW0603100KFKEA	1
3362P-1-303LF	Resistor	R10, R17, R18, R22, R23, R27	3362P_1	3362P-1-303LF	6
0R	Res Thick Film 0402 0 Ohm Epoxy Pad SMD T/R	R12, R13	RESC1005X40N	RC0402JR070RL	2
10k	YAGEO (FH/COMF) RC0603 F, 10k OHM, 0.1W	R14, R15, R16, R19, R20, R21, R24, R25, R26, R29, R30, R31, R34, R35, R36, R39, R40, R41	RESC1608X55N	RC0603FR0710KL	18
1k	0.1/CRCW0603 100 1K1%ET1 e3	R44	RES_CRCW_0603	CRCW06031K00FKEA	1
PTS636SK25FSMTRLFS	PTS636 Series SPST-NO Off-Mom 250 gf Top Actuated SMT Tactile Switch	SW1	PTS636SK25FSMTRLFS	PTS636SK25FSMTRLFS	1
PH1-01-UA	Description	TP1	CONN_PH1-01-UA_ADM	PH1-01-UA	1
STM32C552RET6	Integrated Circuit	U1	QFP50P1200X1200X160-64N	STM32C552RET6	1
AP63203QMU-7	Description	U2	TSOT26_DIO	AP63203QMU-7	1
2171790001	Universal Serial Bus (USB) Shielded I/O Receptacle, Type C, Right-Angle, Top Mount, Surface Mount, Single Row, 16 Circuits, Gold (Au) Plating	U3	2171790001_MOL	2171790001	1
LMR38020FSQDDARQ1	Switching Voltage Regulators Automotive SIMPLESWITCHER power converter 4.2-V to 80-V, 2-A, synchronous buck converter	U4	SOIC127P600X170-9N	LMR38020FSQDDARQ1	1
3296W-1-253LF	TRIMMER,CERMET,RES T,25 KILOHMS,PCB,25 TURNS,PWR-RTG 0.5W	VR1	TRIM_3296W-1-253LF	3296W-1-253LF	1
CS10820-24M	24 MHz +/-15ppm Crystal 6pF 80 Ohms 4-SMD, No Lead , -40C – 125C	X1	CS1082024M	CS10820-24M	1